

*Lambton College at Queen's College of Business, Technology and
Public Safety*

Program: Big Data Analytics (DSMM)

Project:

JobMate: AI-Powered Job and Recruiter Automation
- Milestone 3 Papers Submission For Approval -

Course: Big Data Capstone Project

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Papers about a new technique/technology/system/platform being proposed.

Paper 1:

Title: MetaGPT: Meta Programming for a Multi-Agent Collaborative Framework

Abstract: Recently, remarkable progress has been made on automated problem solving through societies of agents based on large language models (LLMs). Previous LLM-based multi-agent systems can already solve simple dialogue tasks. More complex tasks, however, face challenges through logic inconsistencies due to cascading hallucinations caused by naively chaining LLMs. Here we introduce MetaGPT, an innovative meta-programming framework incorporating efficient human workflows into LLM-based multi-agent collaborations. MetaGPT encodes Standardized Operating Procedures (SOPs) into prompt sequences for more streamlined workflows, thus allowing agents with human-like domain expertise to verify intermediate results and reduce errors. MetaGPT utilizes an assembly line paradigm to assign diverse roles to various agents, efficiently breaking down complex tasks into subtasks involving many agents working together. On collaborative software engineering benchmarks, MetaGPT generates more coherent solutions than previous chat-based multi-agent systems.

Year: 2024

Authors: Sirui Hong, Mingchen Zhuge, Jiaqi Chen, Chenglin Wu, Jürgen Schmidhuber, et al.

Journal/Conference: ICLR 2024 (OpenReview)

Github: <https://github.com/geekan/MetaGPT>

Paper URL: <https://openreview.net/pdf?id=VtmBAGCN7o>

Paper 2:

Title: CAMEL: Communicative Agents for "Mind" Exploration of Large Language Model Society

Abstract: The rapid advancement of chat-based language models has led to remarkable progress in complex task-solving. However, their success heavily relies on human input to guide the conversation, which can be challenging and time-consuming. This paper explores the potential of building scalable techniques to facilitate autonomous cooperation among communicative agents, and provides insight into their "cognitive" processes. To address the challenges of achieving autonomous cooperation, we propose a novel communicative agent framework named role-playing. Our approach involves using inception prompting to guide chat agents toward task completion while maintaining consistency with human intentions. We showcase how role-playing can be used to generate conversational data for studying the

behaviors and capabilities of a society of agents, providing a valuable resource for investigating conversational language models. In particular, we conduct comprehensive studies on instruction-following cooperation in multi-agent settings. Our contributions include introducing a novel communicative agent framework, offering a scalable approach for studying the cooperative behaviors and capabilities of multi-agent systems, and open-sourcing our library to support research on communicative agents and beyond: <https://github.com/camel-ai/camel>.

Year: 2023

Authors: Guohao Li, Hasan Hammoud, Hani Itani, Dmitrii Khizbullin, Bernard Ghanem

Conference: NeurIPS 2023

GitHub: <https://github.com/camel-ai/camel>

Paper URL:

https://proceedings.neurips.cc/paper_files/paper/2023/file/a3621ee907def47c1b952ade25c67698-Paper-Conference.pdf

Paper 3:

Title: Magentic-One: A Generalist Multi-Agent System for Solving Complex Tasks

Abstract: Modern AI agents, driven by advances in large foundation models, promise to enhance our productivity and transform our lives by augmenting our knowledge and capabilities. To achieve this vision, AI agents must effectively plan, perform multi-step reasoning and actions, respond to novel observations, and recover from errors, to successfully complete complex tasks across a wide range of scenarios. In this work, we introduce Magentic-One, a high-performing open-source agentic system for solving such tasks. Magentic-One uses a multi-agent architecture where a lead agent, the Orchestrator, plans, tracks progress, and re-plans to recover from errors. Throughout task execution, the Orchestrator also directs other specialized agents to perform tasks as needed, such as operating a web browser, navigating local files, or writing and executing Python code. Our experiments show that Magentic-One achieves statistically competitive performance to the state-of-the-art on three diverse and challenging agentic benchmarks: GAIA, AssistantBench, and WebArena. Notably, Magentic-One achieves these results without modification to core agent capabilities or to how they collaborate, demonstrating progress towards the vision of generalist agentic systems. Moreover, Magentic-One's modular design allows agents to be added or removed from the team without additional prompt tuning or training, easing development and making it extensible to future scenarios. We provide an open-source implementation of Magentic-One and AutoGenBench, a standalone agentic evaluation tool. AutoGenBench provides built-in controls for repetition and isolation to run agentic

benchmarks where actions may produce side-effects, in a rigorous and contained way. Magentic-One, AutoGenBench and detailed empirical performance evaluations of Magentic-One, including ablations and error analysis are available at <https://aka.ms/magentic-one>.

Year: 2024

Authors: Yi Zhang, Xinlong Wang, Xiaodong Liu, Zihan Liu, Sirui Hong, et al.

Conference: Microsoft Research 2024

GitHub:

<https://github.com/microsoft/autogen/tree/main/python/packages/autogen-magentic-one>

Paper URL: <https://arxiv.org/abs/2411.04468v1>

Papers about reviewing/comparing various technologies/techniques

Paper 1: Beyond Self-Talk: A Communication-Centric Survey of LLM-Based Multi-Agent Systems

Abstract : Large Language Models (LLMs) have recently demonstrated remarkable capabilities in reasoning, planning, and decision-making. Building upon these strengths, researchers have begun incorporating LLMs into multi-agent systems (MAS), where agents collaborate or compete through natural language interactions to tackle tasks beyond the scope of single-agent setups. In this survey, we present a communication-centric perspective on LLM-based multi-agent systems, examining key system-level features such as architecture design and communication goals, as well as internal mechanisms like communication strategies, paradigms, objects and content. We illustrate how these communication elements interplay to enable collective intelligence and flexible collaboration. Furthermore, we discuss prominent challenges, including scalability, security, and multimodal integration, and propose directions for future work to advance research in this emerging domain. Ultimately, this survey serves as a catalyst for further innovation, fostering more robust, scalable, and intelligent multi-agent systems across diverse application domains.

Year: 2025

Authors: Bingyu Yan, Xiaoming Zhang, Litian Zhang, Lian Zhang, Ziyi Zhou, Dezhuang Miao, Chaozhuo Li

Paper Url: <https://arxiv.org/abs/2502.14321>

Paper 2: A survey on LLM-based multi-agent systems: Workflow, Infrastructure, and Challenges

Abstract: The pursuit of more intelligent and credible autonomous systems, akin to human society, has been a long-standing endeavor for humans. Leveraging the exceptional reasoning and planning capabilities of large language models (LLMs), LLM-based agents have been proposed and have achieved remarkable success across a wide array of tasks. Notably, LLM-based multi-agent systems (MAS) are considered a promising pathway towards realizing general artificial intelligence that is equivalent to or surpasses human-level intelligence. In this paper, we present a comprehensive survey of these studies, offering a systematic review of LLM-based MAS. Adhering to the workflow of LLM-based multi-agent systems, we synthesize a general structure encompassing five key components: profile, perception, self action, mutual interaction, and evolution. This unified framework encapsulates much of the previous work in the field. Furthermore, we illuminate the extensive applications of LLM-based MAS in two principal areas: problem-solving and world simulation. Finally, we discuss in detail several contemporary challenges and provide insights into potential future directions in this domain.

Year: 2024

Authors: Xinyi Li, Sai Wang, Siqi Zeng, Yu Wu & Yi Yang

Paper url : <https://link.springer.com/article/10.1007/s44336-024-00009-2>